ECDLP follow-ups: paired factor-base sweep and Kummer addition gate

Reproducible experiment report

21 October 2026

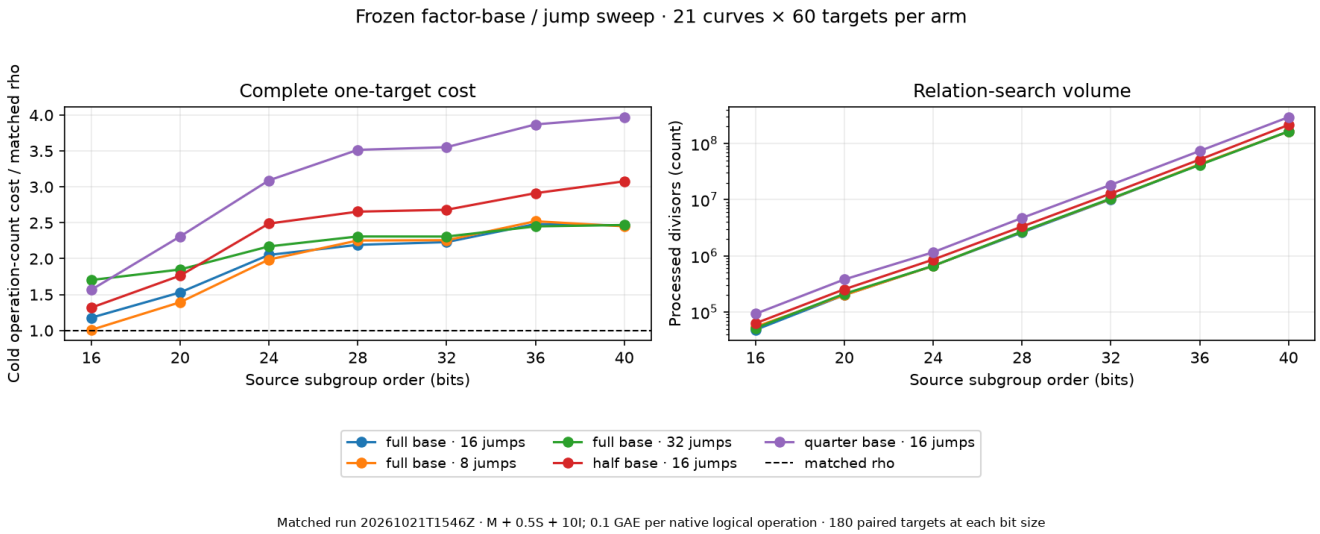
Scope. The frozen protocol covers four leads on 21 exact-order Scholten models, 60 separate targets per model. This report contains completed factor-base and jump experiments and an exact Kummer applicability gate. The complete inversion-free law and full-cohort pencil collector remain open. All costs are cold one-target operation-count prices, not wall-speed claims.

Factor-base and jump sweep

Five arms solved and source-verified all 1,260 targets each, with zero failures or cap exhaustions. All 6,300 same-process rho controls were also source-verified. All 1,260 incumbent count and phase records replay exactly. At $M + 0.5S + 10I$ and 0.1 source-addition equivalents (GAE) per declared native logical operation, matched negation rho totals 1,088,802,370 GAE.

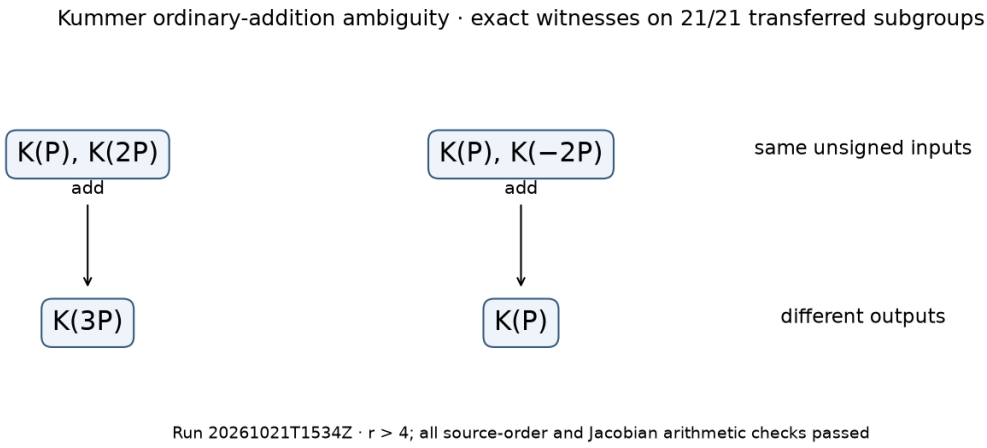
Arm	Divisors processed	Total GAE	/ external	/ rho
Full base, 16 jumps	219,581,773	2,666,291,894	1.000000	2.448830
Full base, 8 jumps	220,092,171	2,664,487,778	0.999323	2.447173
Full base, 32 jumps	219,864,588	2,671,135,032	1.001816	2.453278
Half base, 16 jumps	282,983,105	3,285,713,695	1.232316	3.017732
Quarter base, 16 jumps	392,583,830	4,267,226,320	1.600435	3.919193

The half and quarter bases rejected 35,385,164 and 110,402,348 split relations, respectively, when neither root was in the fixed target-blind base; all rejected candidates were charged. None of the five arms is below rho in the 36 pooled pricing cells. The 8-jump arm is below the 16-jump arm in 15 of 36 cells. Across bit sizes it is below at 16, 20, 24 and 40 bits, and above at 28, 32 and 36 bits. One seed was used per target; no uncertainty interval or isolated wall-time result was measured.



Kummer ordinary-addition gate

Set $P = \tau(G)$ and $Q = 2P$. The unsigned quotient gives $K(Q) = K(-Q)$, but $K(P+Q) = K(3P)$ and $K(P-Q) = K(P)$ differ for order $r > 4$. The source-order, Jacobian-arithmetic and distinct-output checks passed on 21/21 exact-order models. This rules out binary ordinary addition from two unsigned Kummer inputs for the current arbitrary-jump walk. A differential walk with a known difference and sign/lift recovery has not been implemented or priced.



Open requirements and evidence

The published complete inversion-free genus-2 formulas have not been implemented or run on the 1,260 targets. The point-driven pencil collector has a prior 20-target gate but no full-cohort result here. A differential Kummer collector remains unimplemented. These are open requirements, not negative end-to-end verdicts.

The machine record consists of 35 raw matched JSONL arm files, 6,300 paired candidate/rho target rows, the full 36-cell price grid, summary JSON, 21 Kummer witnesses and a checksum manifest under `research/four_followups_20261021/`. The native analyzer pairs workload hashes, seeds, caps and source scalars; the full-base arm reproduces the frozen external-law counts and phases. A separate native validator recomputed 15 headline quantities from raw rows; an archive-input rerun was compared byte-for-byte across all 35 files. The plotting script only renders those recorded values. See https://github.com/aburan28/crypto/pull/2304 for the review branch and https://eprint.iacr.org/2024/623.pdf for the separate complete-law path.